SMART INDIA HACKATHON 2026

Problem Statement ID - SIH26175

Problem Statement Title - Single-View Height Estimation and 3D Flythrough

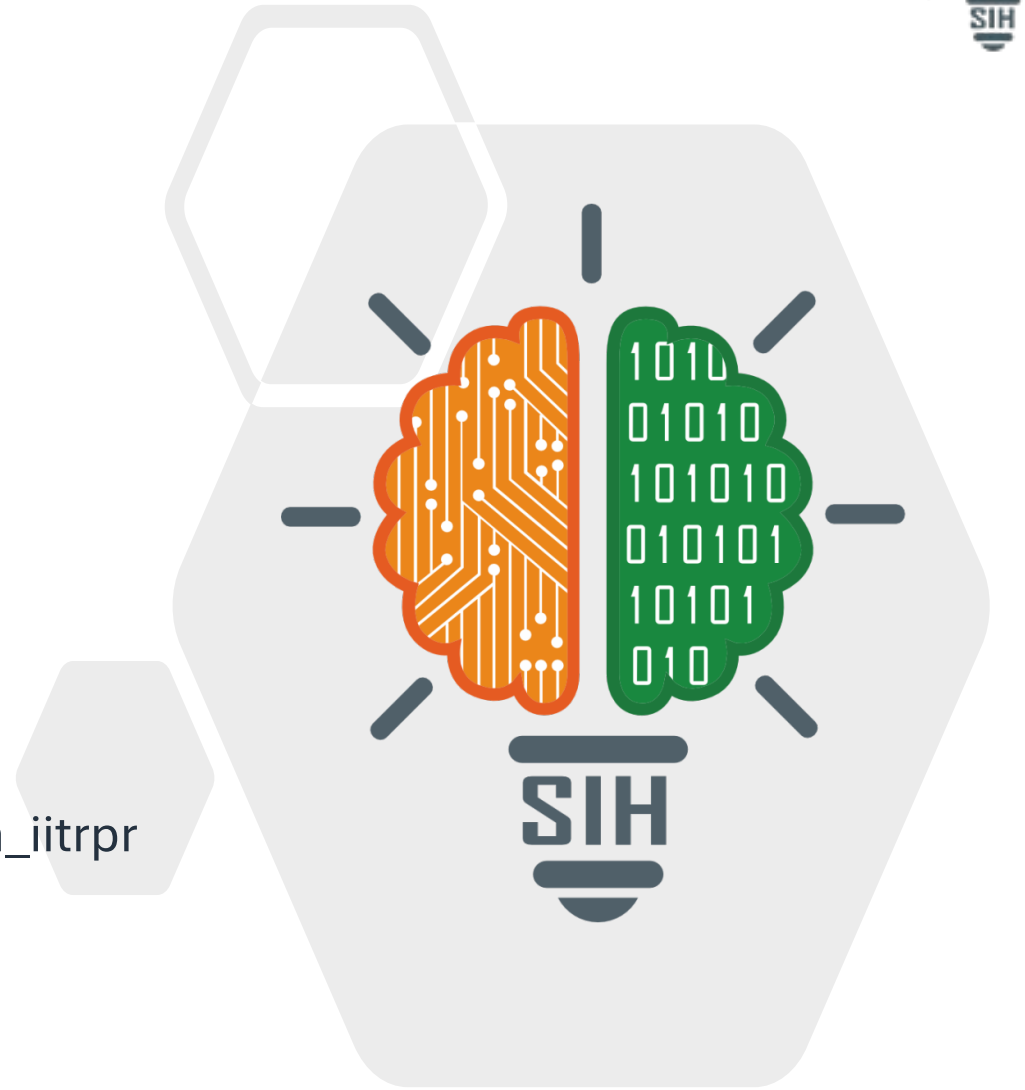
Theme - Disaster Management

PS Category – Software

Team ID - [Registered Team ID]

Team Name (Registered on portal) - TerraVista_iitrpr

Idea Title - DepthWizard



IDEA TITLE

DepthWizard: One Satellite Image → Measured 3D Heights You Can Fly Through and Verify

INPUT	AI HEIGHT	CALIBRATE	VERIFY	FLY & ACT
PNG / JPG / TIFF / GeoTIFF	Height above ground, every pixel	CartoDEM-guided real metres	ICESat-2 laser check + click-to-measure	3D flythrough + flood / slope tools

HOW IT MEETS THE PROBLEM STATEMENT

- ✓ One optical image in: no stereo pair, LiDAR or InSAR needed
- ✓ PNG/JPG → relative DSM · GeoTIFF → absolute DSM in metres
- ✓ Upload → auto-process → fly through the reconstructed 3D terrain
- ✓ Click any building: estimated height ± uncertainty vs reference
- ✓ Complete suite: web app + offline desktop app, source code, docs

WHY DEPTHWIZARD IS DIFFERENT

- Satellite-trained height model; generic depth AI fails on top-down images
- Two accuracy boosters: CartoDEM calibration + ICESat-2 laser correction
- Trust layer: per-pixel uncertainty + live error report in the viewer
- Bird's-eye → street-level view in one click (Sat3DGen)
- Disaster tools on top: flood depth, landslide slopes, affected buildings

INNOVATION & UNIQUENESS

- **Satellite-native height AI**
- **Real metres from CartoDEM**
- **Laser-verified (ICESat-2)**
- **Bird's-eye → street flythrough**

TECHNICAL APPROACH

Pipeline: Pixels → Height → Metres → Verified 3D

INPUT
RGB image (PNG / JPG / TIFF)

PREPROCESS
Read GSD + CRS · overlapping 518 px tiles

AI MODEL
RS3DAda (ViT-L + DPT) · challenger: DINOv3-SAT

HEIGHT
Height above ground + uncertainty map

CALIBRATE
CartoDEM / SRTM prior → metres (Prompt2DEM)

VERIFY
ICESat-2 correction + error report

OUT
DSM + 3D

AI / ML

- RS3DAda (NeurIPS 2024, MIT): released weights
- Challenger: DINOv3-SAT + DPT (CHMv2 init), GAMUS-tuned
- Keep the winner on the GAMUS test set (MAE / RMSE)
- TTA → uncertainty map · Gaussian tile blending
- Distilled small model → ONNX Runtime on CPU

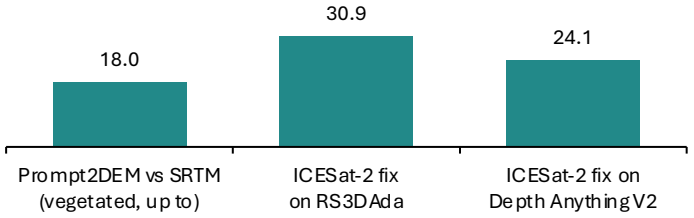
GEOSPATIAL

- GDAL · Rasterio · PyProj; GeoTIFF float32 in / out
- DSM = terrain (CartoDEM / SRTM / GLO-30) + height
- Prompt2DEM (2025): 30 m DEM + RGB → 30 cm heights
- ICESat-2 ATL03 / ATL08: correct bias + report RMSE
- One vertical datum (ICESat-2 is ellipsoidal)

APPLICATION & 3D VIEWER

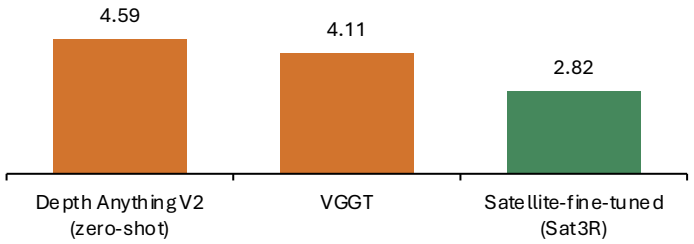
- React + Three.js viewer; GLB mesh + terrain tiles
- Free-fly (WASD), draw-a-path auto-fly, MP4 export
- Street mode: Sat3DGen (ICLR 2026) street mesh
- Click-to-measure: height ± uncertainty anywhere
- FastAPI + CLI; Tauri desktop app, fully offline

Published: calibration cuts error (MAE %)



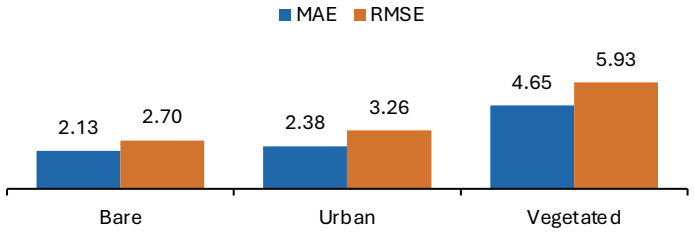
Rafaeli et al., arXiv 2507.09681 (2025) · Song et al., ISPRS J. P&RS vol. 232 (2026)

Published: generic AI vs satellite AI (MAE, m)



Sat3R, arXiv 2605.07264 (2026): DFC2019 multi-view benchmark

Published: error depends on terrain (m)



Prompt2DEM, arXiv 2507.09681 (2025) → we report accuracy per terrain type

Built for both halves of the score: DSM accuracy (RMSE / MAE by terrain) + Rendering & UX (flythrough, measuring, offline)

FEASIBILITY AND VIABILITY

Built on Released Models, Free Data and Offline-Ready Software

FEASIBILITY

- Working baseline on day 1: RS3DAda weights + inference script are public (MIT)
- Training data ready: GAMUS (PS-recommended), DFC2019/US3D, SynRS3D (69k synthetic images)
- Free terrain + truth data: CartoDEM (Bhuvan), SRTM, Copernicus GLO-30, ICESat-2
- Fine-tuning fits one cloud GPU; frozen-backbone option for low budgets

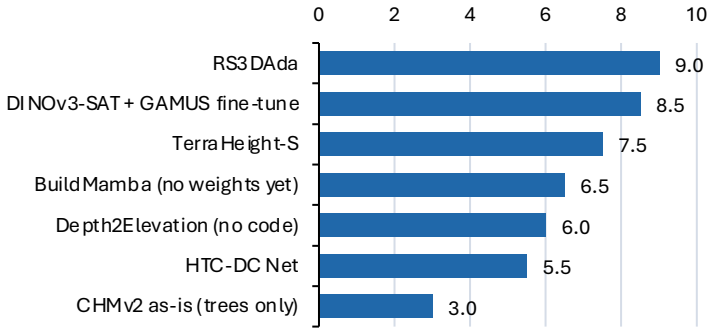
VIABILITY

- Modular: swap the height model without touching calibration or the viewer
- Standard outputs (GeoTIFF DSM, GLB) open in QGIS, ArcGIS and Bhuvan workflows
- Offline: distilled ONNX model + Tauri desktop app + bundled demo data
- License-clean: MIT / Apache / DINOv3 license; no non-commercial weights

CHALLENGE → RISK → MITIGATION

CHALLENGE	RISK	MITIGATION
New image resolution (GSD)	Wrong height scale	Read GSD from metadata · resample · multi-resolution fine-tuning
One image can't see ground level	Heights off in metres	CartoDEM-guided calibration + ICESat-2 correction
Few tall buildings in training data	Tall buildings under-estimated	Oversample tall buildings · report error by height band
Tilted (off-nadir) images	Leaning buildings, shifted roofs	Off-nadir training data (US3D) · warning in UI
No GPU / no internet in the field	Tool unusable	ONNX CPU model · offline desktop app

Model shortlist (team assessment, /10)



Score = fit to PS + public weights + reported accuracy + effort. Multi-view models (need several images) excluded.

IMPACT AND BENEFITS

From One Image to Faster, Verifiable Disaster Decisions

DISASTER MANAGEMENT

Flood-depth simulation · landslide-prone slopes · affected-building counts
Example: raise water to 3 m → see which buildings go under

DEPTHWIZARD

Measured 3D from one image

FROM TERRAIN DATA TO ACTIONABLE INTELLIGENCE

URBAN PLANNING

Building heights and volumes · 3D base maps where no LiDAR exists

REMOTE SENSING (ISRO)

More value from existing optical archives · fresh 3D from every new image, no stereo tasking

1 **HEIGHT MAP** per pixel

2 **VERIFIED 3D** metres + ICESat-2

3 **DECISION** flood / slope

One optical image → measured, checked terrain context

FIELD RESPONSE

Fly-through + street view for team briefings · line-of-sight / viewshed for relief staging

ECONOMIC

Reuse existing images instead of new LiDAR/stereo surveys for every event

SOCIAL

Faster 3D damage picture; works offline in the field

ENVIRONMENTAL

Terrain, slope and canopy context for flood, landslide and forest monitoring

RESEARCH AND REFERENCES

Research Foundation: 2024–2026 Models, Open Data, Independent Validation

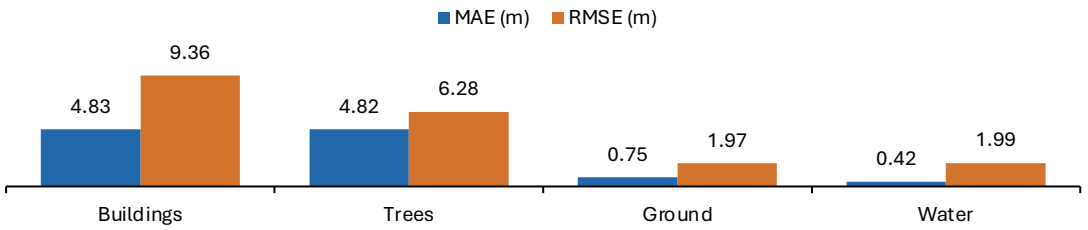
MODELS & METHODS

- RS3DAda / SynRS3D: Song et al., NeurIPS 2024 (primary model)
- DINOv3 (Meta, 2025) + CHMv2 (Scientific Data, 2026): satellite backbone
- Prompt2DEM: Rafaeli et al., 2025, SRTM-prior metric heights (30 m → 30 cm)
- ICESat-2-guided correction: Song, Chen & Yokoya, ISPRS J. 2026
- Sat3DGen (ICLR 2026) · Benchmarks: BuildMamba 2026, Depth2Elevation (TGRS 2025), HTC-DC Net

DATASETS / VALIDATION / OFFICIAL SOURCES

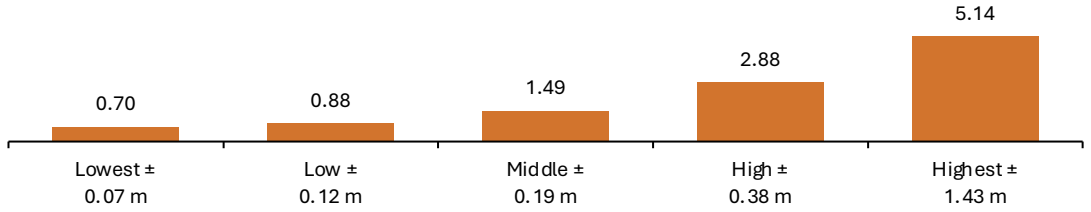
- GAMUS (recommended in PS) · DFC2019 / US3D · SynRS3D
- CartoDEM (ISRO Bhuvan) · SRTM 30 m · Copernicus DEM GLO-30
- ICESat-2 ATL03 / ATL08 (NASA Earthdata)
- ISRO Bhoonidhi / Bhuvan imagery · ISPRS Potsdam / Vaihingen
- Metrics: RMSE and MAE per terrain type and per height band

MEASURED: RS3DAda error vs LiDAR by land cover (m)



Our run: released RS3DAda, 200 DFC2019 test tiles (Jacksonville + Omaha). Official metric MAE 2.22 m / RMSE 3.83 m.

MEASURED: where our ± is high, real error is high (MAE, m)



Same 200 test tiles; pixels split into 5 equal groups by predicted ± (spread of 4 flipped predictions). Rank correlation 0.62.

MODELS	DATASETS	TERRAIN REFERENCE	VALIDATION
RS3DAda + DINOv3	GAMUS + SynRS3D	CartoDEM / SRTM / GLO-30	RMSE / MAE + ICESat-2

Research basis: monocular height estimation • DSM generation • geospatial calibration • ICESat-2 validation • terrain rendering | Evaluation: RMSE / MAE